

# Wearable Smart Insole Based Estimation of Three Dimensional Ground Reaction Forces via Spatio Temporal Attention Fusion

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**Abstract**—Accurate 3D ground reaction force (GRF) estimation traditionally requires expensive, lab-restricted force plates. This study proposes a wearable alternative. We fuse data from smart insoles containing an eighteen-channel pressure array and a six-axis IMU. The model processes these modalities using parallel 1D convolutional encoders. Spatial attention targets specific plantar regions. Temporal self-attention models kinematic sequences within a sliding window. A recurrent layer then fuses these features to regress 3D GRFs. The proposed framework achieved a 3.58% average NRMSE and an  $R^2$  of 0.68. This improves upon a dual-stream attention baseline, which had a 3.90% NRMSE and an  $R^2$  of 0.65. Mediolateral force is highly variable and difficult to infer from pressure sensing alone. For this specific force component, our method reached 1.74% NRMSE and an  $R^2$  of 0.45. This performance exceeds both the baseline and an LSTM reference model.

**Index Terms**—3D GRFs, gait analysis, multimodal fusion, smart insoles, spatio-temporal attention.

## I. INTRODUCTION

3D GRFs are core biomechanical quantities used to characterize human locomotion and functional movement [1], [2]. In this work,  $F_x$  refers to the anterior-posterior GRF (forward and backward),  $F_y$  refers to the mediolateral component (side to side, perpendicular to  $F_x$ ), and  $F_z$  refers to the vertical component. Although force plates can provide highly accurate measurement data, their considerable expense and fixed installation method restrict data acquisition to professional laboratory settings, thereby hindering long-term monitoring in everyday life [2]. Wearable smart insoles that combine plantar pressure sensing with inertial measurement units have therefore become a practical alternative for portable estimation of GRFs [3].

However, performance limitations persist during dynamic motion, particularly when models rely on coarse modal-level fusion, which often fails to adequately capture the coupling between spatial pressure distribution and time-varying kinematics. Existing attention-based fusion methods (including dual-stream designs) typically learn only modal weights, yet fall short of explicitly integrating plantar spatial structure and temporal dependency within a unified representation [4]–[7].

This limitation is especially consequential for the mediolateral force ( $F_y$ ), where signal variability and indirect observability from pressure measurements can degrade estimation accuracy [8].

To address these issues, this paper proposes a spatio-temporal attention fusion framework for wearable 3D GRF estimation using multimodal smart insole measurements. The method integrates spatial attention over plantar pressure channels with temporal self-attention over short kinematic sequences, followed by recurrent fusion for regression of 3D GRFs. The evaluation is conducted against a force plate reference using a standardized movement protocol and includes comparison with a published dual stream attention baseline and a simple LSTM model [4], [9].

## II. METHOD

### A. Experimental Setup

Multimodal data were collected using Tangi0 LTD (TG0) smart insoles that provide an eighteen-channel capacitive plantar pressure array and a six-axis IMU sampled at 100 Hz [4]. Five healthy adults with no reported lower limb pathology performed a standardized sequence of movements comprising squatting, stepping, running, swaying, and jumping. Ground truth 3D GRFs were recorded synchronously using a Kistler 9281CA force plate, and force plate signals were downsampled to 100 Hz to align with the insole sampling rate.

### B. Model Architecture

An overview of the proposed framework is shown in Fig. 1. Raw pressure and IMU signals are preprocessed through denoising and normalization, then segmented into sliding windows of 50 ms. Each modality is processed by a dedicated one-dimensional convolutional encoder to learn modality-specific features. A spatial attention module reweights plantar pressure features to emphasize informative regions that contribute most strongly to force generation, while a temporal attention module based on multi-head self-attention models short-horizon temporal dependencies within each window. The attended features are fused using a gated recurrent unit (GRU), and

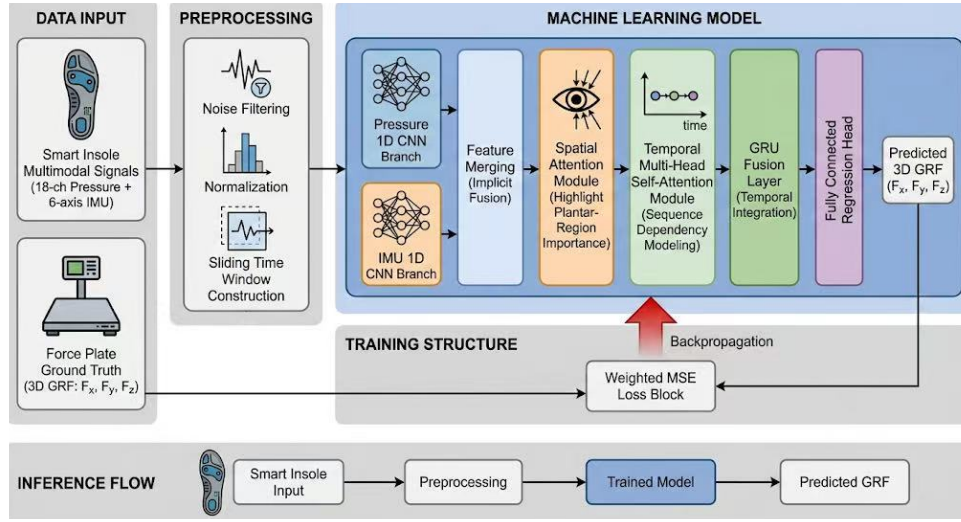


Fig. 1. Overview of the proposed spatio-temporal attention fusion framework for estimation of 3D GRFs from CapSense plantar pressure and IMU measurements.

the output layer regresses the three GRFs. During training, a weighted mean squared error (MSE) objective is used to place increased emphasis on  $F_y$ , reflecting its higher variability and the greater difficulty of inferring lateral dynamics from pressure measurements alone.

### III. RESULTS

Table I summarizes overall and component-level performance. Compared to the dual-stream attention model, the proposed framework reduces average NRMSE from 3.90% to 3.58% (an 8.2% relative reduction) and increases average  $R^2$  from 0.65 to 0.68. Among the force estimations,  $F_y$  exhibits higher variability, which can exacerbate estimation error under wearable sensing constraints [10]. As summarized in Table I, the proposed framework attains 1.74% NRMSE and  $R^2 = 0.45$  for  $F_y$ , exceeding the dual stream attention baseline ( $R^2 = 0.43$ ) and the simple LSTM reference ( $R^2 = 0.21$ ). The improvement is consistent with the design objective of jointly leveraging plantar spatial structure and short-horizon temporal information through the combined attention modules.

### IV. DISCUSSION AND CONCLUSION

In this study, we developed a spatio-temporal attention framework to estimate 3D ground reaction forces (GRF) using smart insoles equipped with both plantar pressure and inertial sensors. Our findings demonstrate that integrating spatial attention across pressure channels with temporal self-attention on kinematic sequences outperforms a traditional dual-stream baseline, yielding particularly strong improvements in medio-lateral force estimations. That said, capturing horizontal forces remains challenging due to the limited sensitivity of capacitive sensors to shear dynamics, and our current findings are constrained by a small sample size [8], [9]. Moving forward, we plan to validate this approach across a more diverse population and explore alternative sensing or modeling configurations to better capture the shear forces necessary for unconstrained, real-world GRF estimation.

TABLE I  
PERFORMANCE COMPARISON (OVERALL AND  $F_y$ ).

| Model             | NRMSE       | $R^2$       | NRMSE ( $F_y$ ) | $R^2$ ( $F_y$ ) |
|-------------------|-------------|-------------|-----------------|-----------------|
| LSTM              | 3.55        | 0.64        | 2.08            | 0.21            |
| Dual-Stream Attn. | 3.90        | 0.65        | 1.77            | 0.43            |
| <b>Proposed</b>   | <b>3.58</b> | <b>0.68</b> | <b>1.74</b>     | <b>0.45</b>     |

### REFERENCES

- [1] T. Sun, D. Li, B. Fan, T. Tan, and P. B. Shull, "Real-time ground reaction force and knee extension moment estimation during drop landings via modular LSTM modeling and wearable IMUs," *IEEE Journal of Biomedical and Health Informatics*, vol. 27, no. 7, pp. 3222–3233, 2023.
- [2] E. Shahabpoor, A. Pavic, and A. Wrigley, "Wearable sensors for ground reaction force measurement in real-life environments: A systematic review," *Sensors*, vol. 20, no. 21, p. 6165, 2020.
- [3] L. A. Cramer, M. A. Wimmer, P. Malloy, J. A. O'Keefe, C. B. Knowlton, and C. Ferrigno, "Validity and reliability of the insole3 instrumented shoe insole for ground reaction force measurement during walking and running," *Sensors*, vol. 22, no. 6, p. 2203, 2022.
- [4] D. Zhang, L. Guo, T. Exell, Y. Huang, and D. Zhou, "Dual-stream attention fusion for 3D GRF estimation with smart insoles," in *Proc. IEEE Int. Conf. Wearable Implantable Body Sensor Networks (BSN)*, 2024, pp. 1–6.
- [5] S. Kim, J. Park, D. Lee, Y. Choi, and H. Kim, "Multi-modal spatio-temporal attention for wearable sensor-based gait analysis," *IEEE Transactions on Biomedical and Health Informatics*, vol. 27, no. 8, pp. 3892–3901, 2023.
- [6] H. Lee, J. Kim, S. Park, S. Lim, and J. Park, "Spatio-temporal attention network for ground reaction force estimation in running," *Journal of Biomechanical Engineering*, vol. 146, no. 5, p. 054502, 2024.
- [7] Y. Chen, C. Yang, H. Zhang, Z. Liu, and J. Wang, "Spatio-temporal graph attention networks for motion capture-based ground reaction force prediction," *Computer Methods in Biomechanics and Biomedical Engineering*, vol. 28, no. 7, pp. 567–578, 2025.
- [8] A. Kammoun, P. Ravier, and O. Buttelli, "Comparison of the accuracy of ground reaction force component estimation between supervised machine learning and deep learning methods using pressure insoles," *Sensors*, vol. 24, no. 16, p. 5318, 2024.
- [9] G. Grouvel, L. Carcreff, F. Moissenet, and S. Armand, "A dataset of asymptomatic human gait and movements obtained from markers, IMUs, insoles and force plates," *Scientific Data*, vol. 10, no. 1, p. 180, 2023.
- [10] T. Yamaguchi, Y. Takahashi, and Y. Sasaki, "Prediction of three-directional ground reaction forces during walking using a shoe sole sensor system and machine learning," *Sensors*, vol. 23, no. 21, p. 8985, 2023.